



Mitoprep Answer Key 10

Physics

1 - B	2 - C	3 - C	4 - A	5 - C	6 - D	7 - A	8 - D	9 - B	10 - C
11 - D	12 - B	13 - C	14 - D	15 - B	16 - D	17 - B	18 - D	19 - D	20 - C
21 - D	22 - B	23 - B	24 - A	25 - A	26 - B	27 - A	28 - B	29 - C	30 - D
31 - B	32 - C	33 - B	34 - D	35 - A	36 - D	37 - C	38 - C	39 - C	40 - A
41 - B	42 - C	43 - B	44 - B	45 - A					

Chemistry

46 - B	47 - C	48 - C	49 - B	50 - B	51 - D	52 - C	53 - D	54 - D	55 - C
56 - C	57 - C	58 - D	59 - B	60 - C	61 - A	62 - B	63 - D	64 - C	65 - A
66 - B	67 - B	68 - C	69 - B	70 - C	71 - D	72 - B	73 - D	74 - A	75 - A
76 - A	77 - B	78 - D	79 - D	80 - D	81 - C	82 - C	83 - A	84 - C	85 - A
86 - A	87 - C	88 - A	89 - B	90 - A					

Biology - (Zoology)

91 - A	92 - C	93 - D	94 - B	95 - B	96 - C	97 - D	98 - B	99 - A	100 - A
101 - A	102 - A	103 - D	104 - A	105 - D	106 - B	107 - C	108 - C	109 - D	110 - A
111 - B	112 - C	113 - B	114 - B	115 - A	116 - B	117 - A	118 - A	119 - C	120 - C
121 - A	122 - A	123 - B	124 - A	125 - D	126 - D	127 - C	128 - A	129 - A	130 - B
131 - B	132 - A	133 - B	134 - C	135 - D					

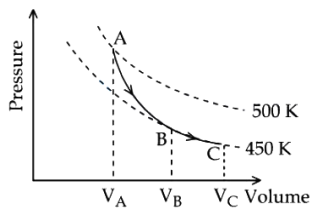
Biology - (Botany)

136 - D	137 - A	138 - D	139 - A	140 - C	141 - D	142 - D	143 - C	144 - A	145 - D
146 - A	147 - B	148 - A	149 - D	150 - A	151 - A	152 - C	153 - C	154 - D	155 - D
156 - B	157 - D	158 - B	159 - C	160 - C	161 - A	162 - B	163 - D	164 - A	165 - B
166 - A	167 - C	168 - C	169 - D	170 - D	171 - D	172 - C	173 - B	174 - B	175 - B
176 - C	177 - D	178 - D	179 - C	180 - B					

Solution

Physics

1.



A poly-atomic molecule ($C_V = 3R, C_P = 4R$, where R is gas constant) goes from phase space point A ($P_A = 10^5 \text{ Pa}$, $V_A = 4 \times 10^{-6} \text{ m}^3$) to point B ($P_B = 5 \times 10^4 \text{ Pa}$, $V_B = 6 \times 10^{-6} \text{ m}^3$) to point C ($P_C = 10^4 \text{ Pa}$, $V_C = 8 \times 10^{-6} \text{ m}^3$). A to B is an adiabatic path and B to C is an isothermal path. The net heat absorbed per unit mole by the system is :

- A) $500R(\ln 3 + \ln 4)$ B) $450R(\ln 4 - \ln 3)$ C) $500R \ln 2$ D) $400R \ln 4$

Solution : (Correct Answer: B)

$$\Delta Q_{AB} = 0 \text{ adiabatic}$$

$$\Delta Q_{BC} = \Delta W_{BC}$$

$$= nRT \ln \left(\frac{V_C}{V_B} \right) = 450R \ln \left(\frac{8 \times 10^{-6}}{6 \times 10^{-6}} \right)$$

$$= 450R \ln \left(\frac{4}{3} \right) = 450R(\ln 4 - \ln 3)$$

$$\therefore \Delta Q = \Delta Q_{AB} + \Delta Q_{BC}$$

$$\Delta Q = 450R(\ln 4 - \ln 3)$$

2. In a stretched string,

- A) Only transverse waves can exist
B) Only longitudinal waves can exist
C) Both transverse and longitudinal waves can exist
D) None of these

Solution : (Correct Answer: C)

(c)

In a solid both transverse and longitudinal waves may pass.

3. For a wave propagating in a medium, identify the property that is independent of the others

- A) velocity
B) wavelength
C) frequency
D) all these depend on each other

Solution : (Correct Answer: C)

In a medium velocity and wavelength are dependent on refractive index of the medium but frequency remains unchanged.

4. A metal rod of 1 m length, is dropped exact vertically on to a hard metal floor. With an oscilloscope, it is determined that the impact produces a longitudinal wave of 1.2 kHz frequency. The speed of sound in the metal rod is

- A) 2400 B) 1800 C) 1200 D) 600

Solution : (Correct Answer: A)

$$v = \lambda f$$

$$v = f_{\min} \lambda_{\max}$$

$$v = f_{\min} \times 2L$$

$$v = 1.2 \times 10^3 \times 2 \times 1$$

$$v = 2400 \text{ m/s}$$

5. A wavelength 0.60 cm is produced in air and it travels at a speed of 300 m/s . It will be an

- A) Audible wave B) Infrasonic wave
C) Ultrasonic wave D) None of the above

Solution : (Correct Answer: C)

$$(c) n = \frac{v}{\lambda} = \frac{300}{0.6 \times 10^{-2}} \text{ Hz}$$

$$= \frac{3}{6} \times 10^4 \text{ Hz} = 50,000 \text{ Hz}$$

==> Wave is ultrasonic.

6. Velocity of sound is maximum in

- A) Air B) Water C) Vacuum D) Steel

Solution : (Correct Answer: D)

(d) Velocity of sound in steel is maximum out of the given materials water and air.

In vacuum sound cannot travel, it's speed is zero.

7. A metal rod of length ' L ' and mass ' m ' is pivoted at one end. A thin disk of mass ' M ' and radius ' R ' ($< L$) is attached at its center to the free end of the rod. Consider two ways the disc is attached: (case A) The disc is not free to rotate about its center and (case B) the disc is free to rotate about its center. The rod-disc system performs SHM in vertical plane after being released from the same displaced position. Which of the following statement(s) is (are) true?

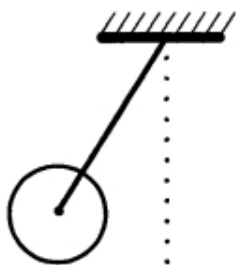
Image

(A) Restoring torque in case A = Restoring torque in case B

(B) Restoring torque in case A < Restoring torque in case B

(C) Angular frequency for case $A >$ Angular frequency for case B .

(D) Angular frequency for case $A <$ Angular frequency for case B .



A) (A, D)

B) (B, D)

C) (C, D)

D) (A, B)

Solution : (Correct Answer: A)

Restoring torque is same in both cases,

$$\tau = \frac{T}{I} = -\omega^2 \theta$$

In case A the moment of Inertia is more as compared to B , So

$$\omega_B > \omega_A$$

A and D are correct answer

8. Which of the following expressions corresponds to simple harmonic motion along a straight line, where x is the displacement and a, b, c are positive constants?

A) $a + bx - cx^2$

B) bx^2

C) $a - bx + cx^2$

D) $-bx$

Solution : (Correct Answer: D)

In linear $S.H.M.$, the restoring force acting on particle should always be proportional to the displacement of the particle and directed towards the equilibrium position.

$$\text{i.e., } F \propto x$$

or $F = -bx$ where b is a positive constant

9. **Assertion** : In SHM , acceleration is always directed towards the mean position.

Reason : In SHM , the body has to stop momentarily at the extreme position and move back to mean position.

A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

C) If the Assertion is correct but Reason is incorrect.

D) If both the Assertion and Reason are incorrect.

Solution : (Correct Answer: B)

SHM is basically to and fro motion about the mean position. So when the body goes away from mean position an acceleration always try to return the

body towards mean position. As the acceleration in SHM is always in opposite phase to that of displacement. The displacement of the particle in SHM at an instant is directed away from the mean position then acceleration at that instant is directed towards the mean position.

10. Which one of the following equations of motion represents simple harmonic motion ?

Where k, k_0, k_1 and a are all positive

A) $\text{Acceleration} = k(x)$

B) $\text{Acceleration} = k(x + a)$

C) $\text{Acceleration} = -k(x + a)$

D) $\text{Acceleration} = -k(x^2)$

Solution : (Correct Answer: C)

Equation of $S.H.M.$ –

$$a = -\frac{d^2x}{dt^2} = -\omega^2 x$$

$$\omega = \sqrt{\frac{k}{m}}$$

–wherein

$$x = A \sin(\omega t + \delta)$$

$$a = -Kx$$

$$x = x + a$$

where $a = -K(x + a)$

In $S.H.M$ acceleration is directly proportional to the displacement from the mean position

Also the acceleration is in the opposite direction of displacement

11. What is constant in $S.H.M.$

A) Restoring force

B) Kinetic energy

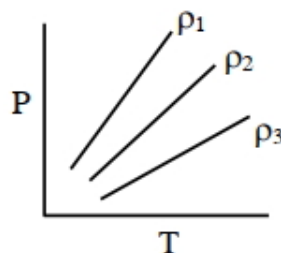
C) Potential energy

D) Periodic time

Solution : (Correct Answer: D)

The only thing that remains constant for one particle performing SHM is its periodic time or simply time period.

12. $P - T$ diagram of an ideal gas having three different densities ρ_1, ρ_2, ρ_3 (in three different cases) is shown in the figure. Which of the following is correct :



A) $\rho_2 < \rho_3$

B) $\rho_1 > \rho_2$

C) $\rho_1 < \rho_2$

D) $\rho_1 = \rho_2 = \rho_3$

Solution : (Correct Answer: B)

For ideal gas

$$PV = nRT$$

$$PV = \frac{m}{M}RT$$

$$P = \left(\frac{M}{V}\right) \frac{RT}{M}$$

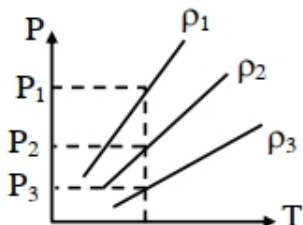
$$P = \frac{\rho RT}{M}$$

(Where m is mass of gas and M is molecular mass of gas)

for same temperature $P_1 > P_2 > P_3$

So $\rho_1 > \rho_2 > \rho_3$

So correct answer is (2)



13. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C . Its density is kg/m^3 ($R = 8.3 \text{ J mol}^{-1} \text{K}^{-1}$)

A) 0.02 B) 0.5 C) 0.2 D) 0.1

Solution : (Correct Answer: C)

For an ideal gas sample

$$\frac{P}{\rho} = \frac{RT}{Mw}$$

$$\rho = \frac{PMw}{RT} = \frac{249 \times 10^3 \times 2 \times 10^{-3}}{8.314 \times 300} = 0.199$$

$$\rho = 0.2 \text{ kg/m}^3$$

14. A given sample of an ideal gas occupies a volume V at a pressure P and absolute temperature T . The mass of each molecule of the gas is m . Which of the following gives the density of the gas ?

A) $\frac{P}{kTV}$ B) mKT
C) $\frac{P}{kT}$ D) $\frac{Pm}{kT}$

Solution : (Correct Answer: D)

$$\text{As } PV = nRT \text{ or } n = \frac{PV}{RT} = \frac{\text{mass}}{\text{molar mass}}$$

... (i)

$$\text{Density, } \rho = \frac{\text{mass}}{\text{volume}} = \frac{(\text{molar mass})P}{RT} = \frac{(mN_A)P}{RT}$$

[From eqn. (i)]

$$\rho = \frac{mP}{kT} \quad (\because R = N_A k)$$

15. The translatory kinetic energy of a gas per gm is

A) $\frac{3}{2} \frac{RT}{N}$ B) $\frac{3}{2} \frac{RT}{M}$
C) $\frac{3}{2} RT$ D) $\frac{3}{2} NKT$

Solution : (Correct Answer: B)

Kinetic energy of 1 gram of gas $\Rightarrow E(\text{transferrential})$

$$= \frac{3}{2} rT = \frac{3}{2} \frac{RT}{M}$$

16. A balloon contains 1500 m^3 of helium at 27°C and 4 atmospheric pressure. The value of helium at -3°C temperature and 2 atmospheric pressure will be m^3

A) 1500 B) 1700 C) 1900 D) 2700

Solution : (Correct Answer: D)

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\Rightarrow V_2 = \frac{P_1 V_1}{T_1} \times \frac{T_2}{P_2}$$

$$\Rightarrow V_2 = \frac{1500 \times 4 \times 270}{300 \times 2} = 2700 \text{ m}^3$$

17. Match List I with List II :

List I	List II
A Isothermal Process	I Work done by the gas decreases internal energy
B Adiabatic Process	II No change in internal energy
C Isochoric Process	III The heat absorbed goes partly to increase internal energy and partly to do work
D Isobaric Process	IV No work is done on or by the gas

Choose the correct answer from the options given below :

A) A – II, B – I, C – III, D – IV

B) A – II, B – I, C – IV, D – III

C) A – I, B – II, C – IV, D – III

D) A – I, B – II, C – III, D – IV

Solution : (Correct Answer: B)

$$\Delta U = nC_v \Delta T$$

For isothermal process T is constant

$$\text{So } \Delta U = 0$$

$$A \rightarrow II$$

Adiabatic process

$$\Delta Q = 0$$

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta U = -\Delta W$$

Work done by gas is positive

So ΔU is negative

$$B \rightarrow I$$

For Isochoric process $\Delta W = 0$

$$C \rightarrow IV$$

For Isobaric process

$$\Delta W = P \Delta V \neq 0$$

$$\Delta U = nC_v \Delta T \neq 0$$

Heat absorbed goes partly to increase internal energy and partly do work.

18. A gas is compressed adiabatically, which one of the following statement is NOT true.

- A) There is no heat supplied to the system
 B) The temperature of the gas increases
 C) The change in the internal energy is equal to the work done on the gas.
 D) There is no change in the internal energy

Solution : (Correct Answer: D)

$$(1) \Delta Q = 0$$

$$(2) \Delta Q = \Delta U + \Delta W$$

$$\Rightarrow \Delta U = -\Delta W$$

adiabatic compression ($V \downarrow$)

$$\Delta W = -ve \Rightarrow \Delta U = +ve$$

$$\Delta U \uparrow \Rightarrow T \uparrow$$

$$\Delta U \neq 0$$

19. During the melting of a slab of ice at 273 K at atmospheric pressure

- A) Positive work is done by ice-water system on the atmosphere
 B) Positive work is done on the ice-water system by the atmosphere
 C) The internal energy of the ice-water system increases
 D) (B) and (C) both

Solution : (Correct Answer: D)

(d) There is a decrease in volume during melting on an ice slab at 273 K .

Therefore, negative work is done by ice-water system on the atmosphere or positive work is done on the ice-water system by the atmosphere.

Hence option (b) is correct. Secondly heat is absorbed during melting (i.e. ΔQ is positive) and as we have seen, work done by ice-water system is negative (ΔW is negative).

Therefore, from first law of thermodynamics $\Delta U = \Delta Q - \Delta W$.

Change in internal energy of ice-water system, ΔU will be positive or internal energy will increase.

20. The state of a thermodynamic system is represented by

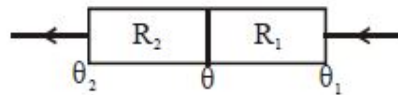
- A) Pressure only
 B) Volume only
 C) Pressure, volume and temperature
 D) Number of moles

Solution : (Correct Answer: C)

(c) The state of a system is defined by ascribing values to a sufficient number of variables. In mechanics, the state of a body is specified by means of a few measurable quantities, such as the position of centre of mass, its velocity and acceleration etc.

In thermodynamics, however we disregard these mechanical coordinates and instead consider the interior of the system, find out directly or indirectly by experiments same measurable quantities upon which the internal state of the system for all thermodynamic purposes, the state of a system can be represented by specifying its pressure p , volume v and the temperature T provided the system is in equilibrium.

21. Consider two insulating sheets with thermal resistances R_1 and R_2 as shown. The temperatures θ is



- A) $\frac{\theta_1 \theta_2 R_1 R_2}{(R_1 + R_2)(\theta_1 + \theta_2)}$
 B) $\frac{\theta_1 R_1 + \theta_2 R_2}{R_1 + R_2}$
 C) $\frac{(\theta_1 + \theta_2)(R_1 R_2)}{R_1^2 + R_2^2}$
 D) $\frac{\theta_1 R_2 + \theta_2 R_1}{R_1 + R_2}$

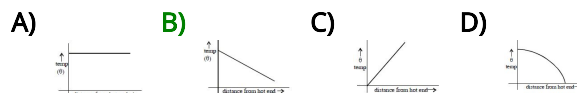
Solution : (Correct Answer: D)

Heat flow in (1) = Heat flow in (2)

$$\frac{\Delta T_1}{R_1} = \frac{\Delta T_2}{R_2}$$

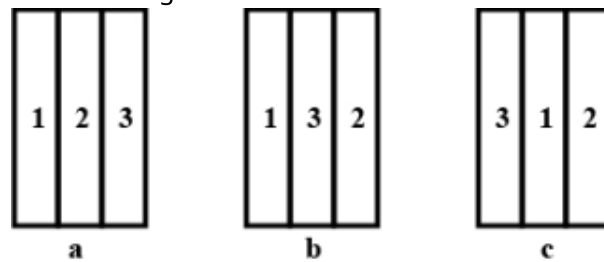
$$\frac{\theta_1 - \theta}{R_1} = \frac{\theta - \theta_2}{R_2} \Rightarrow \theta = \frac{\theta_1 R_2 + \theta_2 R_1}{R_1 + R_2}$$

22. In steady state graph between temp and distance from hot end is



Solution : (Correct Answer: B)

23. Figure shows three different arrangements of materials 1, 2 and 3 to form a wall. Thermal conductivities are $k_1 > k_2 > k_3$. The left side of the wall is 20°C higher than the right side. Temperature difference ΔT across the material 1 has following relation in three cases



- A) $\Delta T_a > \Delta T_b > \Delta T_c$
 B) $\Delta T_a = \Delta T_b = \Delta T_c$
 C) $\Delta T_a = \Delta T_b > \Delta T_c$
 D) $\Delta T_a = \Delta T_b < \Delta T_c$

Solution : (Correct Answer: B)

In all the three cases, temperature difference along the combination is same. Thus, heat current in all the three cases are same $\Rightarrow$ temperature difference across any layer in all the three case is same

24. The ends of two rods of different materials with their thermal conductivities, radii of cross-sections and lengths all are in the ratio 1 : 2 are maintained at the same temperature difference. If the rate of flow of heat in the larger rod is 4 cal/sec, that in the shorter rod in cal/sec will be

A) 1 B) 2 C) 8 D) 16

Solution : (Correct Answer: A)

$$(a) \frac{dQ}{dt} = \frac{K(\pi r^2) d\theta}{dl}$$

$$\Rightarrow \left(\frac{dQ}{dt} \right)_s = \frac{K_s \times r_s^2 \times l_l}{K_l \times r_l^2 \times l_s} = \frac{1}{2} \times \frac{1}{4} \times \frac{2}{1}$$

$$\Rightarrow \left(\frac{dQ}{dt} \right)_s = \frac{\left(\frac{dQ}{dt} \right)_l}{4} = \frac{4}{4} = 1$$

25. Snow is more heat insulating than ice, because

A) Air is filled in porous of snow
 B) Ice is more bad conductor than snow
 C) Air is filled in porous of ice
 D) Density of ice is more

Solution : (Correct Answer: A)

(a) Air is poor conductor of heat.

26. A piece of ice (heat capacity = 2100 J kg⁻¹°C⁻¹ and latent heat = 3.36 × 10⁵ J kg⁻¹) of mass m grams is at -5°C at atmospheric pressure. It is given 420 J of heat so that the ice starts melting. Finally when the ice-water mixture is in equilibrium, it is found that 1 gm of ice has melted. Assuming there is no other heat exchange in the process, the value of m is

A) 7 B) 8 C) 9 D) 5

Solution : (Correct Answer: B)

$$420 = (m \times 2100 \times 5 + 1 \times 3.36 \times 10^5) \times 10^{-3}$$

where m is in gm.

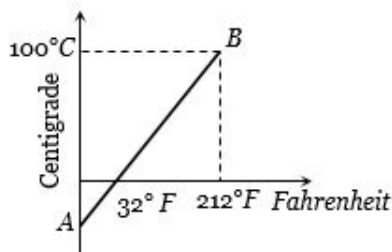
27. If a bimetallic strip is heated, it will

A) Bend towards the metal with lower linear thermal expansion coefficient
 B) Bend towards the metal with higher linear thermal expansion coefficient
 C) Not bend at all
 D) None

Solution : (Correct Answer: A)

Bimetallic strip bends towards the metal with lower liner thermal expansion coefficient on heating.

28. The graph AB shown in figure is a plot of temperature of a body in degree celsius and degree Fahrenheit. Then



A) Slope of line AB is 9/5
 B) Slope of line AB is 5/9
 C) Slope of line AB is 1/9
 D) Slope of line AB is 3/9

Solution : (Correct Answer: B)

(b) Relation between Celsius and Fahrenheit scale of temperature is

$$\frac{C}{5} = \frac{F-32}{9} \Rightarrow C = \frac{5}{9}F - \frac{160}{9}$$

Equating above equation with standard equation of line $y = mx + c$

we get slope of the line AB is $m = \frac{5}{9}$

29. The temperature of a body on Kelvin scale is found to be x K. When it is measured by Fahrenheit thermometer, it is found to be $x^\circ F$, then the value of x is

A) 40 B) 313 C) 574.25 D) 301.25

Solution : (Correct Answer: C)

$$(c) \frac{F-32}{9} = \frac{K-273}{5} \Rightarrow \frac{x-32}{9} = \frac{x-273}{5} \Rightarrow x = 574.25$$

30. On centigrade scale the temperature of a body increases by 30 degrees. The increase in temperature on Fahrenheit scale is °

A) 50 B) 40 C) 30 D) 54

Solution : (Correct Answer: D)

(d)

$$F = \frac{9}{5}C + 32$$

$$\text{So, } F_1 = \frac{9}{5}C_1 + 32 \text{ and } F_2 = \frac{9}{5}C_2 + 32$$

$$F_1 - F_2 = \frac{9}{5}(C_1 - C_2) = \frac{9}{5}(30) = 54^\circ F.$$

31. A small spherical droplet of density d is floating exactly half immersed in a liquid of density ρ and surface tension T . The radius of the droplet is (take note that the surface tension applies an upward force on the droplet)

A) $r = \sqrt{\frac{2T}{3(d+\rho)g}}$ B) $r = \sqrt{\frac{3T}{(2d-\rho)g}}$
 C) $r = \sqrt{\frac{T}{(d-\rho)g}}$ D) $r = \sqrt{\frac{T}{(d+\rho)g}}$

Solution : (Correct Answer: B)

$$B + F = mg$$

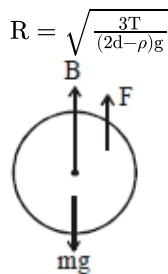
$$B = \left(\frac{2}{3} \pi R^3 \right) \rho g$$

$$F = T(2\pi R)$$

$$m = d \left(\frac{4}{3} \pi R^3 \right)$$

$$\left(\frac{2}{3} \pi R^3 \right) \rho g + T(2\pi R) = d \left(\frac{4}{3} \pi R^3 \right) g$$

$$T(2\pi R) = \left(\frac{2}{3} \pi R^3 \right) g[2d - \rho]$$



32. If T is the surface tension of soap solution, the amount of work done in blowing a soap bubble from a diameter D to $2D$ is

A) $2\pi D^2 T$ B) $4\pi D^2 T$ C) $6\pi D^2 T$ D) $8\pi D^2 T$

Solution : (Correct Answer: C)

Work done to increase the diameter to bubble from d to D

$$W = 2\pi(D^2 - d^2)T = 2\pi[(2D)^2 - (D)^2]T$$

$$= 6\pi D^2 T$$

33. If the surface tension of a soap solution is 0.03 MKS units, then the excess of pressure inside a soap bubble of diameter 6 mm over the atmospheric pressure will be

A) Less than 40 N/m²
 B) Greater than 40 N/m²
 C) Less than 20 N/m²
 D) Greater than 20 N/m²

Solution : (Correct Answer: B)

(b) $\Delta P = \frac{4T}{r} = 40$ N/m²

34. The surface tension for pure water in a capillary tube experiment is

A) $\frac{\rho g}{2hr}$ B) $\frac{2}{hr\rho g}$
 C) $\frac{r\rho g}{2h}$ D) $\frac{hr\rho g}{2}$

Solution : (Correct Answer: D)

(d) $T = \frac{rhdg}{2\cos\theta}$. For pure water $\theta = 0^\circ$ so $T = \frac{rhdg}{2}$

35. A water drop takes the shape of a sphere in a oil while the oil drop spreads in water, because
 (A. F . = adhesive force, C. F . = cohesive force)

A) C. F . for water $>$ A. F . for water and oil
 B) C. F . for oil $>$ A. F . for water and oil
 C) C. F . for oil $<$ A. F . for water and oil
 D) None of the above

Solution : (Correct Answer: A)

A water drop takes the shape of a sphere in a oil while the oil drop spreads in water, because C. F . for water $>$ A. F . for water and oil.

36. Select wrong statement about pressure

A) Pressure is a scalar quantity
 B) Pressure is always compressive in nature
 C) Pressure at a point is same in all directions
 D) None of these

Solution : (Correct Answer: D)

(d)

Pressure is scalar as it is not added vectorially.

Pressure is compressive in nature, it is same in all the directions at a point.

37. By sucking through a straw, a student can reduce the pressure in his lungs to 750 mm of Hg (density = 13.6 gm/cm³). Using the straw, he can drink water from a glass upto a maximum depth of cm

A) 10 B) 75 C) 13.6 D) 1.36

Solution : (Correct Answer: C)

Pressure difference created = 10 mm of Hg

This must be equal to the pressure of water column being created in the straw. If height of water column be h

$$h\rho g = \frac{10}{10} \times 13.6 \times g$$

$$h \times 1 = 13.6 \Rightarrow h = 13.6 \text{ cm.}$$

38. A cubical block of wood of specific gravity 0.5 and a chunk of concrete of specific gravity 2.5 are fastened together. The ratio of the mass of wood to the mass of concrete which makes the combination to float with its entire volume submerged under water is

A) $1/5$ B) $1/3$ C) $3/5$ D) $2/5$

Solution : (Correct Answer: C)

Let the mass of wood and concrete be M and m respectively. Then

$$\frac{M}{0.5} + \frac{m}{2.5} = (M + m)$$

$$\text{Or } M\left(\frac{1}{0.5} - 1\right) = m\left(1 - \frac{1}{2.5}\right)$$

$$\text{or } M = m\left(\frac{1.5}{2.5}\right) = m\left(\frac{3}{5}\right)$$

$$\therefore \frac{M}{m} = \frac{3}{5}$$

39. During blood transfusion the needle is inserted in a vein where the gauge pressure is 2000 Pa. At what height (in m) must the blood container be placed so that blood may just enter the vein? Density of whole blood, $\rho = 1.06 \times 10^3$ kgm⁻³

A) 0.1 B) 0.3 C) 0.2 D) 0.4

Solution : (Correct Answer: C)

Gauge pressure, $P = 2000$ Pa Density of whole blood, $\rho = 1.06 \times 10^3$ kgm⁻³

Acceleration due to gravity, $g = 9.8$ m/s² Height of the blood container = h Pressure of the blood container, $P = h\rho g \therefore h = \frac{P}{\rho g}$

$$= \frac{2000}{1.06 \times 10^3 \times 9.8}$$

$$= 0.1925m$$

The blood may enter the vein if the blood container is kept at a height greater than 0.1925 m, i.e., about $0.2m$

40. Two substances of densities ρ_1 and ρ_2 are mixed in equal volume and the relative density of mixture is 4. When they are mixed in equal masses, the relative density of the mixture is 3. The values of ρ_1 and ρ_2 are

- A) $\rho_1 = 6$ and $\rho_2 = 2$
 B) $\rho_1 = 3$ and $\rho_2 = 5$
 C) $\rho_1 = 12$ and $\rho_2 = 4$
 D) None of these

Solution : (Correct Answer: A)

(a) When substances are mixed in equal volume then density = $\frac{\rho_1 + \rho_2}{2} = 4 \Rightarrow \rho_1 + \rho_2 = 8 \dots\dots(i)$

When substances are mixed in equal masses then density = $\frac{2\rho_1\rho_2}{\rho_1 + \rho_2} = 3$

$$\Rightarrow 2\rho_1\rho_2 = 3(\rho_1 + \rho_2) \dots\dots(ii)$$

By solving (i) and (ii) we get $\rho_1 = 6$ and $\rho_2 = 2$.

41. **Assertion** : Solids are least compressible and gases are most compressible.

Reason : solids have definite shape and volume but gases do not have either definite shape or definite volume.

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
 B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
 C) If the Assertion is correct but Reason is incorrect.
 D) If both the Assertion and Reason are incorrect.

Solution : (Correct Answer: B)

The incompressibility of solids is primarily due to the tight coupling between the neighbouring atoms. Molecules in gases are very poorly coupled to their neighbours.

42. One end of a uniform wire of length L and of weight W is attached rigidly to a point in the roof and a weight W_1 is suspended from its lower end. If S is the area of cross-section of the wire, the stress in the wire at a height $3L/4$ from its lower end is

- A) $\frac{W_1}{S}$
 B) $\frac{W_1 + (W/4)}{S}$
 C) $\frac{W_1 + (3W/4)}{S}$
 D) $\frac{W_1 + W}{S}$

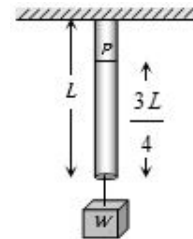
Solution : (Correct Answer: C)

(c) Total force at height $3L/4$ from its lower end

= Weight suspended + Weight of $3/4$ of the chain

$$= W_1 + (3W/4)$$

$$\text{Hence stress} = \frac{W_1 + (3W/4)}{S}$$



43. Modulus of rigidity of diamond is

- A) Too less
 B) Greater than all matters
 C) Less than all matters
 D) Zero

Solution : (Correct Answer: B)

(b)

44. The length of an elastic string is a metre when the longitudinal tension is 4 N and b metre when the longitudinal tension is 5 N . The length of the string in metre when the longitudinal tension is 9 N is

- A) $a - b$
 B) $5b - 4a$
 C) $2b - \frac{1}{4}a$
 D) $4a - 3b$

Solution : (Correct Answer: B)

(b) Let L is the original length of the wire and K is force constant of wire.

Final length = initial length + elongation

$$L' = L + \frac{F}{K}$$

For first condition $a = L + \frac{4}{K} \dots(i)$

For second condition $b = L + \frac{5}{K} \dots(ii)$

By solving (i) and (ii) equation we get

$$L = 5a - 4b \text{ and } K = \frac{1}{b-a}$$

Now when the longitudinal tension is 9 N ,

$$\text{length of the string} = L + \frac{9}{K} = 5a - 4b + 9(b - a)x = 5b - 4a.$$

45. A weight of 200 kg is suspended by vertical wire of length 600.5 cm . The area of cross-section of wire is 1 mm^2 . When the load is removed, the wire contracts by 0.5 cm . The Young's modulus of the material of wire will be

- A) $2.35 \times 10^{12}\text{ N/m}^2$
 B) $1.35 \times 10^{10}\text{ N/m}^2$
 C) $13.5 \times 10^{11}\text{ N/m}^2$
 D) $23.5 \times 10^9\text{ N/m}^2$

Solution : (Correct Answer: A)

(a) $F = 2000\text{ N}$, $L = 6\text{ m}$, $l = 0.5\text{ cm}$, $A = 10^{-6}\text{ m}^2$

$$Y = \frac{FL}{Al} = \frac{2000 \times 6}{10^{-6} \times 0.5 \times 10^{-2}} = 2.35 \times 10^{12}\text{ N/m}^2$$

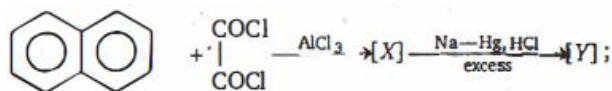
Chemistry

46. Which isomer of xylene can give three different monochloroderivatives?

- A) *o*-xylene
 B) *m*-xylene
 C) *p*-xylene
 D) xylene cannot give a monochloro derivative

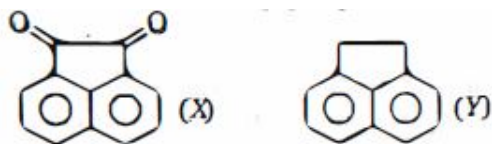
Solution : (Correct Answer: B)

47. Product Y is



- A)
- B)
- C)
- D)

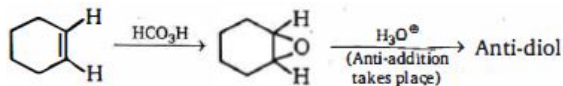
Solution : (Correct Answer: C)



48. Trans-cyclohexane-1,2-diol can be obtained by the reaction of cyclohexene with

- A) $KMnO_4$
 B) OsO_4
 C) peroxy formic acid / H_3O^+
 D) SeO_2

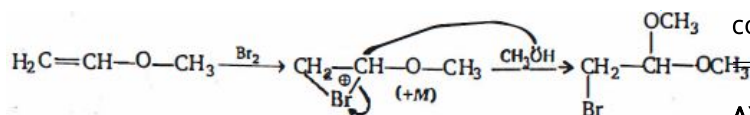
Solution : (Correct Answer: C)



49. Methyl vinyl ether, $H_2C = CH - OCH_3$, reacts with Br_2/CH_3OH . If methanol is reacting as water would, and if this reaction follows a typical mechanism of electrophilic addition, what would be the expected product?

- A)
- B)
- C)
- D)

Solution : (Correct Answer: B)



50. If *p*-methoxy toluene is nitrated, the major product is:

- A)
- B)
- C)
- D) No reaction

Solution : (Correct Answer: B)

51. Which of the following has maximum stability

- A) CH_3^+
 B) $CH_3 - \overset{+}{C} - H$
 C) $CH_3 - \overset{+}{C} - CH_3$
 D) $CH_3 - \overset{+}{C} - CH_3$

Solution : (Correct Answer: D)

Stability of carbonium ion or carbocation depends on the adjacent alkyl groups. Alkyl groups exhibit electron releasing +I effect and hyperconjugative effect which stabilizes the positive charge. CH_3^+ methyl carbonium ion does not have both of these effects, therefore it is the least stable.

Therefore $CH_3 - \overset{+}{C} - CH_3$ have maximum stability.

52. Which represents an intermediate formed in the reaction of toluene and chlorine at elevated temperature in sunlight?

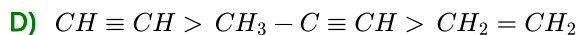
- A)
- B)
- C)
- D)

Solution : (Correct Answer: C)

(c) Free radical mechanism.

53. The correct order for acid strength of compounds $CH \equiv CH$, $CH_3 - C \equiv CH$, $CH_2 = CH_2$ is as follows

- A) $CH \equiv CH > CH_2 = CH_2 > CH_3 - C \equiv CH$
 B) $CH_3 - C \equiv CH > CH \equiv CH > CH_2 = CH_2$
 C) $CH_3 - C \equiv CH > CH_2 = CH_2 > CH \equiv CH$



Solution : (Correct Answer: D)

Acidic strength $\propto$ Stability of conjugate base $E. N.$

$\rightarrow sp \text{ carbon} > sp^2 \text{ carbon} > sp^3 \text{ carbon}$

54. Which of the following molecules have non-zero dipole moments ?

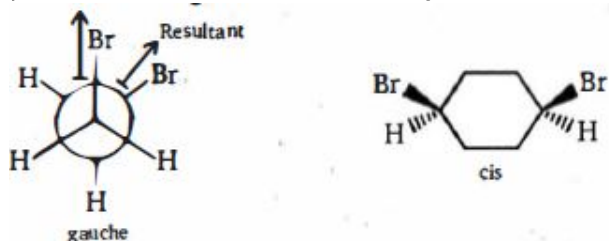
- (I) gauche conformation of 1, 2 -dibromoethane
 (II) anti conformation of 1, 2 -dibromoethane
 (III) trans- 1, 4 -dibromocyclohexane
 (IV) cis- 1, 4 -dibromocyclohexane
 (V) tetrabromomethane
 (VI) 1, 1 -dibromocyclohexane

- A) I and II B) I and IV
 C) II and V D) I, IV and VI

Solution : (Correct Answer: D)

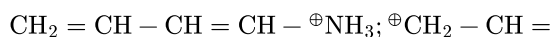
(d) Non-zero dipole has following molecules

Dibromocyclohexane $\rightarrow$ stereo chemistry and position are not mentioned so may-be cis.



55. $CH_2 = CH - CH = CH - ^\oplus NH_3$;
 (I)
 $^\oplus CH_2 - CH = CH - ^\ominus CH - ^\oplus NH_3$
 (II)
 $^\oplus CH_2 - CH = CH - CH = NH_3$
 (III)

Which of these structures is not a valid canonical structure ?



- A) I B) II
 C) III D) none of these

Solution : (Correct Answer: C)

(c) Maximum co-valency of N is 4.

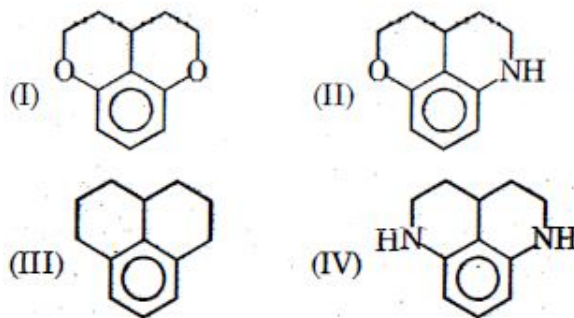
56. Which of the following compounds will be most easily attacked by an electrophile

- A) B)
 C) D)

Solution : (Correct Answer: C)

(c) Phenol is most easily attacked by an electrophile because presence of $-OH$ group increases electron density at $o-$ and $p-$ positions.

57. In above compounds, decreasing order of rate of electrophilic aromatic substitution will be ?



- A) $I > II > III > IV$
 B) $III > IV > II > I$
 C) $IV > II > I > III$
 D) $IV > I > II > III$

Solution : (Correct Answer: C)

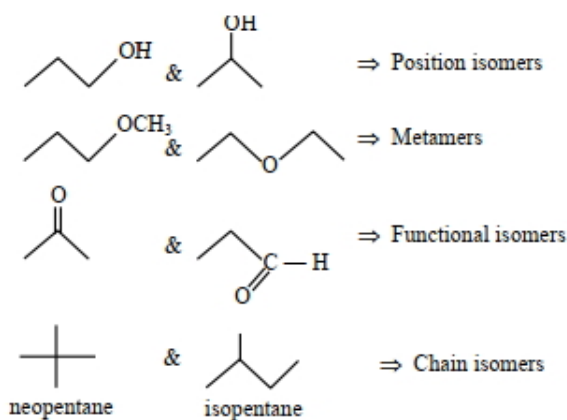
58. Match List - I with List - II

List - I (Pair of Compounds)	List - II (Isomerism)
An-propanol and Isopropanol	(I) Metamerism
BMethoxypropane and ethoxyethane	(II) Chain Isomerism
CPropanone and propanal	(III) Position Isomerism
DNeopentane and Isopentane	(IV) Functional Isomerism

Choose the correct answer from the options given below:

- A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
 B) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
 C) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)
 D) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

Solution : (Correct Answer: D)

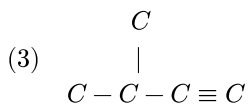


59. The number of possible structural alkynes with molecular formula, C_5H_8 is

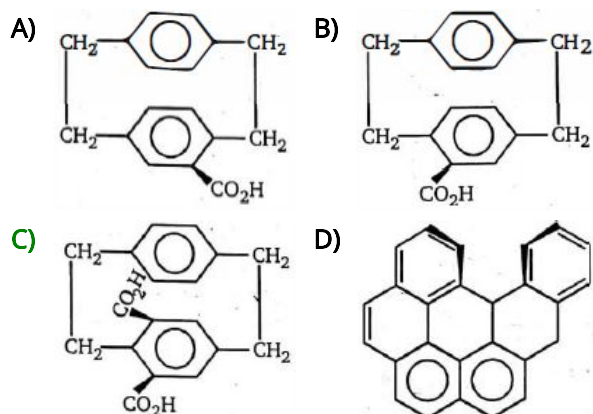
- A) 2 B) 3 C) 4 D) 5

Solution : (Correct Answer: B)



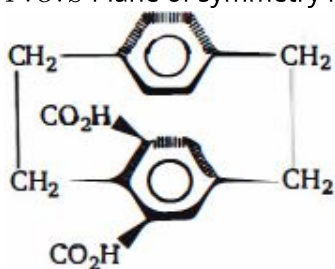


60. Which of the following compound is achiral ?



Solution : (Correct Answer: C)

now, plane passing through square will give P. O. S Plane of symmetry is present.



61. Predict stereochemistry of product when *d* and *l*-amine reacts with *l*-acid

- A) Diastereomers B) Meso
C) Racemic D) Pure Enantiomer

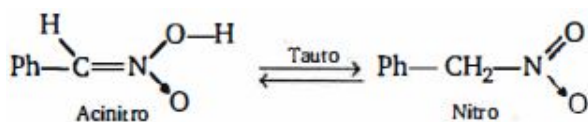
Solution : (Correct Answer: A)

(a) When racemic mixture reacts with optically active compound, it gives diastereoisomeric mixture.

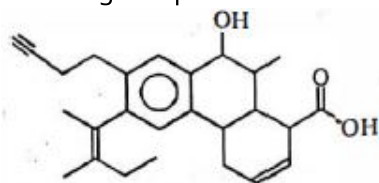
62. $Ph - CH = NO_2H \xrightarrow[3\text{days}]{\text{isomerises}} (x)$, Isomer (x) is (50–50%)

- A) $Ph - NO - CH_2OH$
B) $Ph - CH_2 - NO_2$
C) $Ph - NH - CO_2H$
D) None

Solution : (Correct Answer: B)



63. How many degrees of unsaturation are there the following compound ?

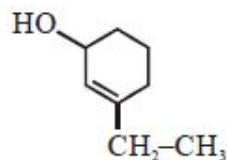


- A) 6 B) 7 C) 10 D) 11

Solution : (Correct Answer: D)

(d) Degree of unsaturation = $(C + 1) - \left(\frac{H + X - N}{2}\right)$

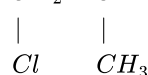
64. The IUPAC name of following compound is



- A) 1-Ethyl cyclohex-1-en-3-ol
B) 1-Ethyl-3-hydroxy cyclohex-1-ene
C) 3-Ethyl cyclohex-2-en-1-ol
D) 2-Ethyl-6-hydroxy cyclohexene

Solution : (Correct Answer: C)

65. The IUPAC name of $CH_2 - C = CH - CH_2OH$ is



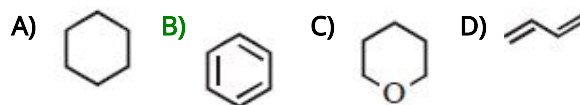
- A) 4-Chloro-3-methylbut-2-en-1-ol
B) 1-Chloro-2-methylbut-2-en-4-ol
C) 4-Chloro-1-hydroxy-3-methylbut, 2-ene
D) 1-Chloro-4-hydroxy-2-methylbut-2-ene

Solution : (Correct Answer: A)

Priority of -OH group is greater than halide group. Numbering will be started from right end.

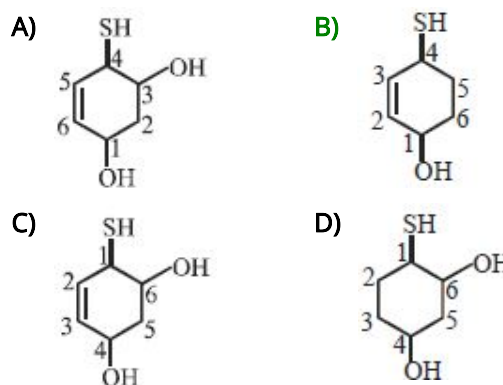
∴ IUPAC name: 4-chloro-3-methylbut-2-en-1-ol.

66. Identify the compound which is homocyclic, aromatic, and unsaturated ?



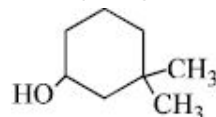
Solution : (Correct Answer: B)

67. Select the structure with correct numbering for IUPAC name of the compound



Solution : (Correct Answer: B)

68. The IUPAC name of the given compound is :



- A) 1,1-dimethyl-3-hydroxy cyclohexane

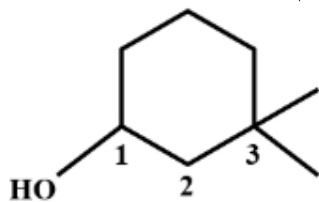
B) 3,3-dimethyl-1-hydroxy cyclohexane

C) 3,3-dimethyl-1-cyclohexanol

D) 1,1-dimethyl-3-cyclohexanol

Solution : (Correct Answer: C)

Hence, IUPAC name is 3,3-dimethyl-1-cyclohexanol



So, number of moles of sodium sulphite that reacts with one mole of $\text{KMnO}_4 = \frac{5}{2}$

73. The correct set of oxidation number of N in NH_4NO_2 is

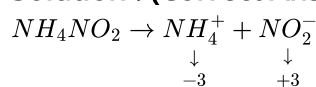
A) -3, +5

B) +5, -3

C) -3, -3

D) -3, +3

Solution : (Correct Answer: D)



74. Assign A, B, C, D from given type of reaction. $\text{I}_2 + \text{NaOH} \rightarrow \text{NaI} + \text{NaOI}$

A) for disproportionation reaction.

B) for comproportionation reaction.

C) for either intermolecular redox reaction or displacement reaction.

D) for either thermal combination redox reaction or thermal decomposition redox reaction.

Solution : (Correct Answer: A)

75. In a chemical reaction, a reductant

A) Loses electron (s)

B) Gains electron (s)

C) loses and gains electrons both

D) No change in electrons

Solution : (Correct Answer: A)

Reductant loses electron (s)

76. What is the molar solubility of :- Ag_2CO_3 ($K_{sp} = 4 \times 10^{-13}$) in 0.1 M Na_2CO_3 solution?

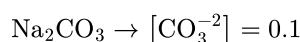
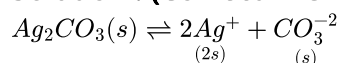
A) 10^{-6}

B) 10^{-7}

C) 2×10^{-6}

D) 2×10^{-7}

Solution : (Correct Answer: A)



$$K_{sp} = [\text{Ag}^+]^2 \times [\text{CO}_3^{2-}]$$

$$4 \times 10^{-13} = (2s)^2 \times (s + 0.1)$$

$\downarrow$

Neglected

$$s = 10^{-6}$$

77. Would gaseous HCl be considered as an Arrhenius acid

A) Yes

B) No

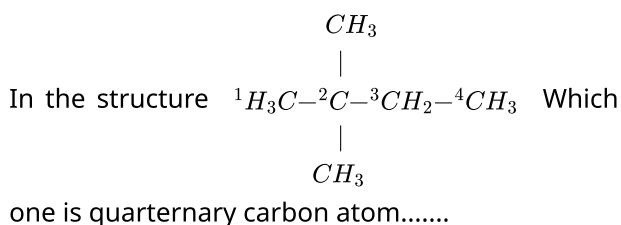
C) Not known

D) Gaseous HCl does not exist

Solution : (Correct Answer: B)

(b) Gaseous HCl does not give H^+ but liquid HCl gives H^+ in aqueous solution there for gaseous HCl is not a Arrhenius acid due to covalent bonding in gaseous condition.

69.



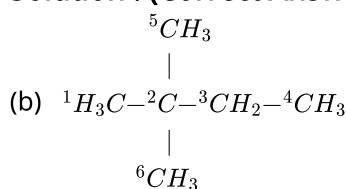
A) 1

B) 2

C) 3

D) 5

Solution : (Correct Answer: B)



C - 2 is quaternary carbon because it is attached to 4 other carbon atoms.

70. The compound which could not act both as oxidising as well as reducing agent is

A) SO_2

B) MnO_2

C) Al_2O_3

D) CrO

Solution : (Correct Answer: C)

(c) Al_2O_3 could not act as a oxidising and reducing agent.

71. In the following reaction, $3\text{Br}_2 + 6\text{CO}_3^{2-} + 3\text{H}_2\text{O} \rightarrow 5\text{Br}^- + \text{BrO}_3^- + 6\text{HCO}_3^-$

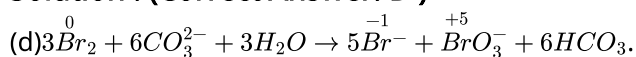
A) Bromine is oxidised and carbonate is reduced

B) Bromine is reduced and water is oxidised

C) Bromine is neither reduced nor oxidised

D) Bromine is both reduced and oxidised

Solution : (Correct Answer: D)



In this reaction bromine is oxidised as well as reduced.

72. The number of moles of sodium sulphite (Na_2SO_3) needed to react with one mole of KMnO_4 in acidic solution

A) 0.8

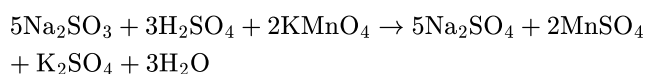
B) 2.5

C) 1

D) 0.6

Solution : (Correct Answer: B)

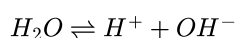
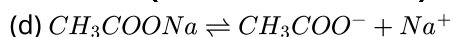
Reaction:



78. An example for a strong electrolyte is

- A) Urea
- B) Ammonium hydroxide
- C) Sugar
- D) Sodium acetate

Solution : (Correct Answer: D)



79. Which of the following substance is an electrolyte

- A) Chloroform
- B) Benzene
- C) Toluene
- D) Magnesium chloride

Solution : (Correct Answer: D)

(d) $MgCl_2 \rightleftharpoons Mg^{++} + 2Cl^-$ in aqueous solution it is dissociated into ions.

80. Which is generally true about ionic compounds

- A) Have low boiling point
- B) Have low melting point
- C) Soluble in non polar solvents
- D) Conduct electricity in the fused state

Solution : (Correct Answer: D)

(d) Generally ionic compound are conduct electricity in fused state.

81. The addition of a polar solvent to a solid electrolyte results in

- A) Polarization
- B) Association
- C) Ionization
- D) Electron transfer

Solution : (Correct Answer: C)

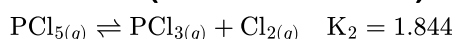
(c) Polar solvent facilitate ionisation of strong electrolytes due to dipole-ion attraction.

82. $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ $K_c = 1.844$

3.0 moles of PCl_5 is introduced in a 1 L closed reaction vessel at 380 K. The number of moles of PCl_5 at equilibrium is $\dots \times 10^{-3}$. (Round off to the Nearest Integer)

- A) 1500
- B) 1292
- C) 1400
- D) 5123

Solution : (Correct Answer: C)



t = 0 3 moles

t = ∞ x x

$$\Rightarrow \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{x^2}{3-x} = 1.844$$

$$\Rightarrow x^2 + 1.844 - 5.532 = 0$$

$$\Rightarrow x = \frac{-1.844 + \sqrt{(1.844)^2 + 4 \times 5.532}}{2}$$

$$\cong 1.604$$

$$\Rightarrow \text{Moles of } PCl_5 = 3 - 1.604 \cong 1.396$$

83. The yield of product in the reaction, $A_2(g) + 2B(g) \rightleftharpoons C(g) + Q$ kJ would be higher at

- A) low temperature and high pressure
- B) high temperature and high pressure
- C) low temperature and low pressure
- D) high temperature and low pressure

Solution : (Correct Answer: A)

Most favourable conditions of temperature and pressure for greater yield of C are high pressure and low temperature.

High pressure is maintained because the moles of the product is less than the moles of reactants.

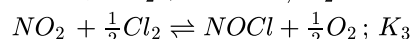
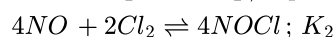
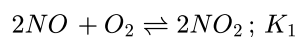
$$\Delta n_g = 1 - (1 + 2) = -2, \Delta n_g < 0$$

When pressure is increased, the equilibrium will shift to the product side (which contains less number of moles of gaseous species). This nullifies the effect of the increase in pressure. Hence, more and more product will be formed.

According to Le Chatelier's principle, when a system at equilibrium is disturbed, the position of the equilibrium shifts in a direction so that the effect of the change is nullified.

Low temperature favours exothermic reaction. Heat is evolved during the reaction (positive value of enthalpy change).

84. For the reactions

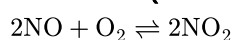


where K_1, K_2, K_3 are equilibrium constants then K_3^2 equal to

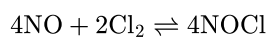
$$A) \sqrt{K_2/K_1} \quad B) \sqrt{K_1 K_2}$$

$$C) \sqrt{K_2}/K_1 \quad D) \frac{1}{K_1 K_2}$$

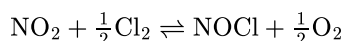
Solution : (Correct Answer: C)



$$K_1 = \frac{[NO_2]^2}{[NO]^2[O_2]} \dots (i)$$



$$K_2 = \frac{[NOCl]^4}{[NO]^4[Cl_2]^2} \dots (ii)$$



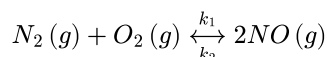
$$K_3 = \frac{[NOCl][O_2]^{1/2}}{[NO_2][Cl_2]^{1/2}} \dots (iii)$$

$$\text{divide } K_2 \text{ by square of } K_1 \frac{[NOCl]^4}{[NO]^4[Cl_2]^2} \times \frac{[NO]^4[O_2]^2}{[NO_2]^4}$$

$$= \frac{[NOCl]^4[O_2]^2}{[NO_2]^4[Cl_2]^2}$$

$$K_2 \times \frac{1}{K_1^2} = K_3^4 \Rightarrow \frac{K_2}{K_1^2} = K_3^4$$

85. For the reversible reaction in equilibrium



$C_0 = Ce^{-2.1 \times 10^{-3} t}$ for the forward reaction and
 $C'_0 = C'e^{-4.2 \times 10^{-4} t}$ for the backward reaction,
hence K_c for the above equilibrium is

- A)** 5.0 **B)** 2.0 **C)** 0.5 **D)** 2

Solution : (Correct Answer: A)

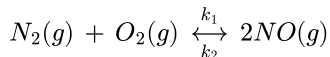
$$C = C_0 e^{-kt}$$

$$\therefore k_f = 2.1 \times 10^{-3}$$

$$k_b = 4.2 \times 10^{-4}$$

$$K_c = \frac{k_f}{k_b} = 5.0$$

- 86.** For the reversible reaction in equilibrium



$C_0 = Ce^{-2.1 \times 10^{-3} t}$ for the forward reaction and $C'_0 = C'e^{-4.2 \times 10^{-4} t}$ for the backward reaction, hence K_c for the above equilibrium is :-

- A)** 5.0 **B)** 2.0 **C)** 0.5 **D)** 0.2

Solution : (Correct Answer: A)

$$C = C_0 e^{-kt}$$

$$k_f = 2.1 \times 10^{-3}$$

$$k_b = 4.2 \times 10^{-4}$$

$$K_c = \frac{k_f}{k_b} = 5.0$$

- 87.** Total number of moles for the reaction $2HI \rightleftharpoons H_2 + I_2$. if α is degree of dissociation is

- A)** 1 **B)** $2 - \alpha$
C) 2 **D)** $1 - \alpha$

Solution : (Correct Answer: C)

$$(c) \quad \text{Initial} \quad \begin{matrix} 2HI \\ 2 \end{matrix} \rightleftharpoons \begin{matrix} H_2 \\ 0 \end{matrix} + \begin{matrix} I_2 \\ 0 \end{matrix} \quad \text{At equilibrium} \quad 2 - \alpha \quad \frac{\alpha}{2} \quad \frac{\alpha}{2}$$

$$= 2 - \alpha + \frac{\alpha}{2} + \frac{\alpha}{2} = 2.$$

- 88.** According to Le-chatelier's principle, an increase in the temperature of the following reaction will
 $N_2 + O_2 \rightleftharpoons 2NO - 43,200 \text{ kcal}$

- A)** Increase the yield of NO
B) Decrease the yield of NO
C) Not effect the yield of NO
D) Not help the reaction to proceed in forward direction

Solution : (Correct Answer: A)

(a) Being endothermic, the forward reaction is favoured by high temperature.

- 89.** The unit of equilibrium constant K for the reaction $A + B \rightleftharpoons C$ would be

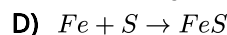
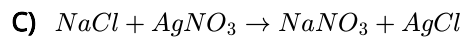
- A)** mol litre^{-1} **B)** litre mol^{-1}
C) mol litre **D)** Dimensionless

Solution : (Correct Answer: B)

unit of eq constant reaction is $\text{litre} / \text{mol}$

- 90.** Which of the following reactions is reversible

- A)** $H_2 + I_2 \rightarrow 2HI$
B) $H_2SO_4 + Ba(OH)_2 \rightarrow BaSO_4 + 2H_2O$



Solution : (Correct Answer: A)

It's Obvious

Biology - (Zoology)

- 91.** Match List I with List II

List I	List II
A. Pons	I. Provides additional space for Neurons, regulates posture and balance.
B. Hypothalamus	II. Controls respiration and gastric secretions.
C. Medulla	III. Connects different regions of the brain.
D. Cerebellum	IV. Neuro secretory cells

Choose the correct answer from the options given below :

A) $A - III, B - IV, C - II, D - I$

B) $A - I, B - III, C - II, D - IV$

C) $A - II, B - I, C - III, D - IV$

D) $A - II, B - III, C - I, D - IV$

Solution : (Correct Answer: A)

The correct answer is option (1) as

A. Pons	—	Part of hindbrain, it connects different regions of the brain.
B. Hypothalamus	—	Also have neuro secretory cells which secrete hormones.
C. Medulla	—	Part of hindbrain which controls respiration and gastric secretions.
D. Cerebellum	—	Part of hindbrain with convoluted surface which provides additional space for neurons, also regulates posture and balance.

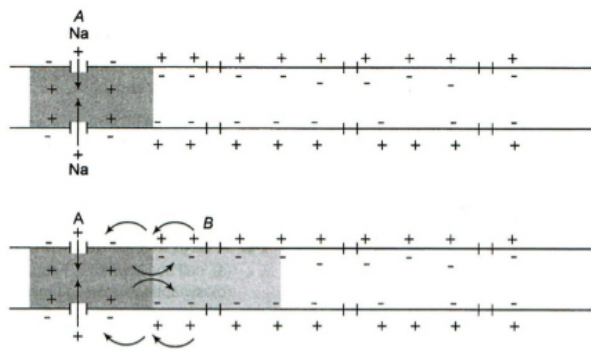
- 92.** Given is the diagrammatic representation of impulse conduction through an axon (at points A and B). View the diagram and arrange the steps of impulse conduction

I. The polarity of the membrane at site A is reversed and depolarized, i.e., the outer surface becomes negatively charged and the innerside becomes positively charged, generating nerve impulse

II. A stimulus causes disturbance to the membrane at site of A nerve fibre resulting in leakage of Na^{+ii} ions inside the nerve fibre

III. On the outer surface, current flows from site B to site A to complete the circuit of current flow. Hence, the polarity at the site is reversed, and an action potential is generated at site B. The impulse (action potential) generated at site A arrives at site B. The sequence is repeated along the length of the axon and consequently the impulse is conducted

IV. Immediately ahead, the axon (e.g., site B) membrane has a positive charge on the outer surface and a negative charge on its inner surface. As a result, a current flows on the inner surface from site A to site B



- A) $I \rightarrow II \rightarrow IV \rightarrow III$
 B) $II \rightarrow I \rightarrow III \rightarrow IV$
 C) $II \rightarrow I \rightarrow IV \rightarrow III$
 D) $I \rightarrow IV \rightarrow III \rightarrow II$

Solution : (Correct Answer: C)

$II \rightarrow I \rightarrow IV \rightarrow III$

93. The system, responsible for providing an organized network of point to point connections for a quick coordination, is called

- A) Endocrine system B) Circulatory system
 C) Digestive system D) Neural system

Solution : (Correct Answer: D)

The system, which is responsible for providing an organized network of point to point connection for a quick coordination is called neural system. This system is made up of highly specialized cells called neurons, which detects the stimuli throughout the body and transmit it to the brain

94. In which of the following very simple neural organization is found ?

- A) Insect B) Hydra C) Frog D) Human

Solution : (Correct Answer: B)

95. Select the correct arrangement of neural organization, according to the increasing degree of complexity.

- A) Lower invertebrates $\rightarrow$ Vertebrates $\rightarrow$ Insects
 B) Lower invertebrates $\rightarrow$ Insects $\rightarrow$ Vertebrates
 C) Vertebrates $\rightarrow$ Insects $\rightarrow$ Lower vertebrates
 D) Vertebrates $\rightarrow$ Lower invertebrates $\rightarrow$ Insects

Solution : (Correct Answer: B)

96. It is not found in or a part of myelinated nerve fibres

- A) Schwann cell B) Node of ranvier
 C) Nissl's granules D) Synaptic knob

Solution : (Correct Answer: C)

97. Which of the following part contains centers, control respiration cardiovascular reflexes and gastric secretions

- A) Cerebral aqueduct B) Pons
 C) Cerebellum D) Medulla

Solution : (Correct Answer: D)

98. The hormones that initiates ejection of milk, stimulates milk production and growth of ovarian follicles are respectively known as

- A) PRL, OT and LH
 B) OT, PRL and FSH
 C) LH, PRL and FSH
 D) PRH, OT and LH

Solution : (Correct Answer: B)

Pituitary gland or the master endocrine gland secretes various hormones controlling the functioning of other endocrine glands. Follicle stimulating hormone (gonadotrophic hormone) is a proteinaceous hormone secreted by gonadotrophs or gynandrotroph cells of anterior pituitary. It stimulates spermatogenesis in testis and maturation of Graafian/ovarian follicles in ovaries with secretion of oestrogen hormone in females. Prolactin/lactogenic or luteotrophic hormone (PRL, LTH) is a proteinaceous hormone secreted by lactotroph cells of anterior pituitary. It stimulates development of mammary glands (in pregnancy) and lactation (after delivery). It is also called as maternity hormone. Oxytocin (OT =pitocin) is a peptide hormone secreted by posterior hormone secreted by posterior pituitary. It functions in vasodilation and in stimulating uterine contraction during delivery. Hence, it is known as birth hormone. Its other function is initiating ejection of milk, so called as milk ejection hormone.

99. Endocrine glands are

- A) Ductless glands whose secretions pour directly into blood
 B) Have ducts and pour their secretions into blood directly
 C) Have ducts which straightway pour secretions into target organs
 D) All of the above

Solution : (Correct Answer: A)

Endocrine glands (ductless glands) or gland of internal secretion have no ducts and their secretion get absorbed into the immediate surrounding

blood circulation to reach the specific organ to initiate a particular metabolic change.

100. Hypersecretion of growth hormone in adults does not cause further increase in height, because

A) epiphyseal plates close after adolescence
B) bones lose their sensitivity to growth hormone in adults
C) muscle fibres do not grow in size after birth
D) growth hormone becomes inactive in adults.

Solution : (Correct Answer: A)

(a) : Epiphyseal plate is a hyaline cartilage plate in the metaphysis at each end of long bone. It is part of long bone where new bone growth takes place. In adults, elevated levels of GH results in acromegaly where no increase in height occurs because of ossified epiphyseal plate.

101. Endocrine glands produce or Action of endocrine glands is mediated through

A) Hormones B) Enzymes
C) Minerals D) Vitamins

Solution : (Correct Answer: A)

It's Obvious

102. Which of the following cell does not secrete hormone

A) Kupffer cell
B) Leydig cell
C) Lutein cell
D) Parafollicular cells of thyroid

Solution : (Correct Answer: A)

(a) Kupffer's cells occur in the lining of liver sinusoids and help in phagocytosis.

103. Body co-ordination is exhibited by

A) Blood vascular system
B) Nervous system
C) Endocrine system
D) Nervous and endocrine system

Solution : (Correct Answer: D)

It's Obvious

104. Endocrine glands

A) Do not possess ducts
B) Sometimes do not have ducts
C) Pour their secretion into blood through ducts
D) Always have ducts

Solution : (Correct Answer: A)

(a) Endocrine gland is ductless gland these secretion flow directly into blood stream.

105. Select the correct matching of the type of the joint with the example in human skeletal system:

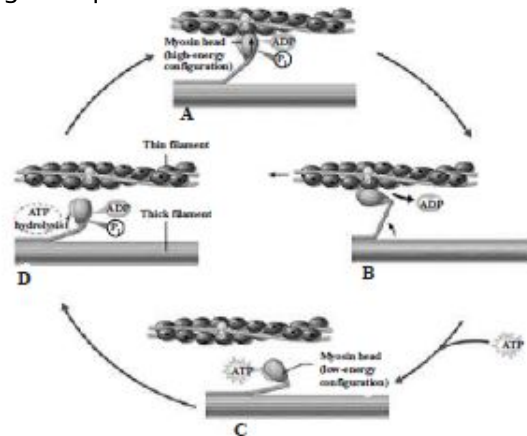
A) Type of joint – Cartilaginous joint, Example – between frontal and parietal

- B) Type of joint – Pivot joint, Example – between third and fourth cervical vertebrae
C) Type of joint – Hinge joint, Example – between humerus and pectoral girdle
D) Type of joint – Gliding joint, Example – between carpals

Solution : (Correct Answer: D)

A gliding joint is a common type of synovial joint formed between bones that meet at flat or nearly flat articular surfaces. Gliding joints allow the bones to glide past one another in any direction along the plane of the joint - up and down, left and right, and diagonally. Many gliding joints are formed in the appendicular skeleton between the carpal bones of the wrist; between the carpals and the metacarpals of the palm; between the tarsal bones of the ankle; and between the tarsals and the metatarsals of the foot.

106. The given figure represents the cross bridge cycle in skeletal muscle. What does the step B in the figure represents ?



- A) Attachment of myosin head to actin forming cross bridge.
B) Release of phosphate. Myosin changes shape to pull actin
C) Attachment of new ATP to myosin head. The cross bridge detaches
D) Splitting of ATP into ADP and Pi. Myosin cocks into its high energy conformation.

Solution : (Correct Answer: B)

Step A: Attachment of myosin head to actin forming cross bridge.

Step B: Release of phosphate. Myosin changes shape to pull actin.

Step C: Attachment of new ATP to myosin head. The cross bridge detaches.

Step D: Splitting of ATP into ADP and Pi. Myosin cocks into its high energy conformation

107. In the rib cage formation which is not used ?

A) Sternum B) Ribs
C) Lungs D) Thoracic vertebra

Solution : (Correct Answer: C)

108. Muscles of the heart are

- A) Voluntary striated
- B) Voluntary smooth
- C) Involuntary striated
- D) Involuntary smooth

Solution : (Correct Answer: C)

It's obvious.

109. In human body which one of the following is anatomically correct?

- A) Collar bones –3 pairs
- B) Kidney –2 pairs
- C) Cranial nerves –10 pairs
- D) Floating ribs –2 pairs

Solution : (Correct Answer: D)

110. The cup-shaped structure of pelvic girdle, the acetabulum in man is formed by

- A) Ilium, ischium and pubis
- B) Ilium, ischium and cotyloid
- C) Ilium and ischium
- D) Ilium and cotyloid

Solution : (Correct Answer: A)

(a) Pelvic girdle consists of three separate bones, ilium, ischium and pubis. On its outer surface it has a deep depression called the acetabulum.

111. The pectoral and pelvic girdles and the bones of limb form

- A) Axial skeleton
- B) Appendicular skeleton
- C) Visceral skeleton
- D) Outer skeleton

Solution : (Correct Answer: B)

(b) Appendicular skeleton is situated at the lateral sides which actually extend outwards from the principal axis. It consists of pectoral and pelvic girdle and bones of arms and legs.

112. Nitrogenous waste is excreted in the form of pellet or paste by :

- A) Salamandra
- B) Hippocampus
- C) Pavo
- D) Ornithorhynchus

Solution : (Correct Answer: C)

113. Which of the following pairs is wrong?

- A) Uricotelic -Birds
- B) Ureotelic -Insects
- C) Ammonotelic -Bony fishes
- D) Ureotelic -Elephant

Solution : (Correct Answer: B)

Insects are uricotelic

114. Human beings are

- A) Uricotelic
- B) Ureotelic
- C) Ammonotelic
- D) Both (b) and (c)

Solution : (Correct Answer: B)

Ureotelic -Human beings

Uricotelic -Insects, birds

Ammonotelic -Tadpole

115. An advantage of excreting nitrogenous wastes in the form of uric acid is that

- A) Uric acid can be excreted in almost solid form
- B) The formation of uric acid requires a great deal of energy
- C) Uric acid is the first metabolic breakdown products of acids
- D) Uric acid may be excreted through the lungs

Solution : (Correct Answer: A)

(a) The animals which excrete mainly uric acid are uricotelic. Uric acid can be stored or excreted in crystalline form.

116. In the kidney, glucose is mainly absorbed in

- A) Loop of Henle
- B) Proximal convoluted tubules
- C) Distal convoluted tubules
- D) Bowman's capsule

Solution : (Correct Answer: B)

(b) Complete reabsorption by active transport takes places for glucose in the proximal convoluted tubules.

117. Uricotelic mode of passing out nitrogenous wastes is found in

- A) Reptiles and Birds
- B) Birds and Marine fishes
- C) Aquatic amphibians and Reptiles
- D) Insects and amphibians

Solution : (Correct Answer: A)

118. The most abundant, harmful and universal waste product of metabolism is

- A) CO_2
- B) Uric acid
- C) H_2O
- D) None of these

Solution : (Correct Answer: A)

(a) CO_2 and H_2O are the end product of complete aerobic oxidation of food through kreb's cycle and electron transport chain (E.T.C.) CO_2 is most abundant, harmful and universal waste product of metabolism.

119. The opening of pulmonary vein is without valve because

- A) It is a very small aperture
- B) It has low blood pressure
- C) Its opening is oblique
- D) None of these

Solution : (Correct Answer: C)

Veins have valves which prevent backward flow of blood except pulmonary vein because it has oblique opening.

120. Sequence of electrical impulse in heart beat is

- A) AV node → pacemaker → auricles → ventricles
- B) Ventricle → pacemaker → AV node → auricle
- C) Pacemaker → atria → AV node → ventricle
- D) Pacemaker → AV node → atria → ventricle

Solution : (Correct Answer: C)

Sequence of electrical impulse in heart is

Sinoauricular node (Pacemaker of heart)

Atrioventricular node (AV node)

Bundle of His

1

Ventricles

Purkinje fibre

121. Grouping of ABO blood is based on the

- A) Surface antigens present on RBCs
- B) Surface lipids present on the cell membrane
- C) Nature of all constituents
- D) Nature of RBC and WBC

Solution : (Correct Answer: A)

ABO blood grouping is based on the presence or absence of the surface antigens, A and B on RBCs

122. Fibrins are formed by the conversion of inactive fibrinogens in the plasma by the enzyme.....

- A) Thrombin
- B) Fibrinogen
- C) Prothrombin
- D) Thrombokinase

Solution : (Correct Answer: A)

123. Match the following.

Column – I	Column – II
(1) Neutrophil	(a) Inflammation
(2) Eosinophil	(b) 6 – 8 %
(3) Basophil	(c) Phagocytes
(4) Monocytes	(d) Resists infections

- A) (1 – d), (2 – c), (3 – a), (4 – b)
- B) (1 – c), (2 – d), (3 – a), (4 – b)
- C) (1 – a), (2 – c), (3 – b), (4 – d)
- D) (1 – b), (2 – a), (3 – c), (4 – d)

Solution : (Correct Answer: B)

124. Which of the following is a cell fragment?

- A) Blood platelets
- B) Bone cells
- C) Lymphocytes
- D) Leucocytes

Solution : (Correct Answer: A)

Blood Platelets occur only in mammals. They are non-nucleated and colourless. They bud off from the megakaryocytic cells of red bone marrow. That's why they are called blood platelets or cell fragments. They have thromboplastin necessary for blood clotting

125. Mark the odd one

- A) Monocytes
- B) Lymphocyte
- C) Neutrophils
- D) Erythrocytes

Solution : (Correct Answer: D)

It's Obvious

126. Match List I with List II :

List I	List II
A. Expiratory capacity	– Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
B. Functional residual capacity	– Tidal volume + Expiratory reserve volume
C. Vital capacity	– Tidal volume + Inspiratory reserve volume
D. Inspiratory capacity	– Expiratory reserve volume + Residual volume

Choose the correct answer from the options given below :

- A) A – III, B – II, C – IV, D – I
- B) A – II, B – I, C – IV, D – III
- C) A – I, B – III, C – II, D – IV
- D) A – II, B – IV, C – I, D – III

Solution : (Correct Answer: D)

Expiratory capacity = Tidal volume + Expiratory reserve volume

Functional residual capacity = Expiratory reserve volume + Residual volume

Vital capacity = Expiratory reserve volume + Tidal volume + Inspiratory reserve volume

Inspiratory capacity = Tidal volume + Inspiratory reserve volume

127. Process of exchange of O_2 from the atmosphere with ...A... produced by the cells is called ...B..., which is commonly known as ...C...

Choose the appropriate options for the blanks A, B and C to complete the given NCERT statement

- A) A – H_2O , B – breathing, C – respiration
- B) A – O_2 , B – breathing, C – respiration
- C) A – CO_2 , B – breathing, C – respiration
- D) A – NO_2 , B – breathing, C – respiration

Solution : (Correct Answer: C)

128. What is correct at alveoli ?

- A) High PO_2
- B) High PCO_2
- C) High H^+ concentration
- D) Higher temperature

Solution : (Correct Answer: A)

129. CO_2 is carried by haemoglobin as carbamino haemoglobin. This binding CO_2 with Hb is related to the partial pressure of

- A) O_2, CO_2 B) Hb, RBC
C) O_2, Hb D) CO_2, H_2O

Solution : (Correct Answer: A)

130. Find out the false statement.

- A) The overall exchange of gases between atmosphere, blood and cell is called respiration.
B) Excretory system and respiratory system involve in exchange of respiratory gases.
C) Cells need a continuous supply of oxygen to carry out activities for their existence.
D) Cellular respiration is the part of catabolism.

Solution : (Correct Answer: B)

131. The normal shape of diaphragm is

- A) Flat B) Dome-like
C) Spherical D) Cone-like

Solution : (Correct Answer: B)

It's obvious.

132. Oxy-haemoglobin dissociates into oxygen and deoxy-haemoglobin at

- A) Low O_2 pressure in tissue
B) High O_2 pressure in tissue
C) Equal O_2 pressure inside and outside tissue
D) All times irrespective of O_2 pressure

Solution : (Correct Answer: A)

(a) More active tissue has much lower P_{O_2} . Therefore much more oxygen is released from oxyhaemoglobin in the active tissue.

133. In expiration condition, diaphragm becomes

- A) Circular B) Relaxed
C) Fully contracted D) Expanded

Solution : (Correct Answer: B)

(b) During expiration diaphragm relaxes and becomes dome-shaped thereby reducing intrathoracic volume to pass out the air from body.

134. Which of the following statement is correct

- A) Breathing refers to chest movement and respiration to the another mechanism
B) Breathing means getting in of CO_2 in the lungs and O_2 out from the lungs
C) Respiration means the oxidation of food material
D) Breathing and respiration both are same

Solution : (Correct Answer: C)

It's obvious.

135. In which of the following animals, respiration occurs with out any respiratory organ

- A) Fish B) Frog
C) Cockroach D) Earthworm

Solution : (Correct Answer: D)

(d) Earthworm respire, but has no respiratory organs, exchange of gases takes place through moist skin.

Biology - (Botany)

136. In coleoptile tissue, auxin is

- A) Not transported because it is used where it is made
B) Transported by diffusion
C) Transported from the base to tip by osmosis
D) Produced by growing apices of stem, which migrate to the region of its action

Solution : (Correct Answer: D)

Auxin is produced by growing apical part of the plant, i.e., apices of stems and roots. Then, it goes to the lateral parts (basipetal) and causes, the apical (root and shoot) parts of the plant to elongate

137. Which of the following movements in plants is due to the increased concentration of auxin?

- A) Movement of shoot towards the source of light
B) Nyctinasty
C) Movement of sunflower towards sun
D) All of the above

Solution : (Correct Answer: A)

Auxin concentration increases in shaded area. i.e., auxins are collected in the opposite side of light. Increased auxin concentration is stimulatory for shoot growth and for this reason, shaded side shows more growth than the lighted side. Hence, bending of shoot takes place towards light.

138. I. Increased vacuolation

II. Cell enlargement

III. New cell wall deposition

Which of the above are the characteristics of phase of elongation?

Choose the correct option accordingly

- A) I and II
B) II and III
C) I and III
D) I, II and III

Solution : (Correct Answer: D)

Characters of phase of elongation phase are

- (i) cell enlargement
(ii) new cell wall deposition
(iii) increased vacuolation

139. Which hormone inhibit morphogenesis

- A) ABA B) 2,4 - D
C) Jasmonic acid D) IBA

Solution : (Correct Answer: A)

ABA (abscisic acid) is a hormone that generally acts as a growth inhibitor. It inhibits the process of morphogenesis.

140. In order to increase the yield of sugarcane crop, which of the following plant growth regulators should be sprayed?

A) Ethylene B) Auxins
C) Gibberellins D) Cytokinins

Solution : (Correct Answer: C)

141. Increase in the girth of plant is known as/done by

A) Primary growth
B) Apical meristem
C) Intercalary meristem
D) Secondary growth

Solution : (Correct Answer: D)

Increase in girth is secondary growth.

Increase in height is primary growth.

142. Find out the correct statement(s).

(a) Growth in plants is internal/intrinsic and open ended
(b) Formation of cellular materials is called real or protoplasmic growth
(c) Plant growth is diffused only during the early embryonic stage

A) Only (a) & (b)
B) (b) only
C) Only (b) & (c)
D) (a), (b) & (c)

Solution : (Correct Answer: D)

143. $CH_2 = CH_2$ is mainly responsible for

A) Formation of internode
B) Formation of nodes
C) Ripening of fruits
D) Formation of internodes

Solution : (Correct Answer: C)

Ethylene is helpful in ripening of fruits.

144. If an etiolated stem could be first saturated with auxin by spraying and then exposed to a streak of light from one side, it will

A) Bend towards the light
B) Bend away from the light
C) Grow straight upwards
D) Be prevented from growing

Solution : (Correct Answer: A)

Auxin causes phototropic response.

145. Which of the following structures show unlimited growth in plants?

A) Leaves B) Flowers C) Fruits D) Roots

Solution : (Correct Answer: D)

Roots, stems show continuous, indeterminate growth due to presence of apical meristem.
Leaves, fruits, flowers show closed, determinate growth due to intercalary meristem.

146. Cytokinins are formed in

A) Roots B) Leaves C) Fruits D) Stems

Solution : (Correct Answer: A)

(a) Cytokinin concentration are highest in meristematic regions and area of continuous growth such as root, young leaves, developing fruits and seed.

147. If a planter is interested in obtaining a good crop of tea leaves from a single plant, he should

A) Feed auxin to the plant through soil
B) Remove the apical bud of the main shoot and the branches
C) Cut off the tip of the plant and then apply auxins to the cut end
D) Supply auxin from the tip of the plant as well as through roots

Solution : (Correct Answer: B)

It's obvious.

148. Which two factors primarily affect the developmental phase of growth of plants

A) Light and temperature
B) Rainfall and temperature
C) Light and wind
D) Temperature and relative humidity

Solution : (Correct Answer: A)

(a) Differential length of light and temperature influence the developmental phase of plant. Photoperiodism and vernalization are the two important factors responsible for the flowering of plants.

149. Where would you look for active cell division in plant

A) In the pith cells
B) In the cells of cortex
C) In the internodal region
D) At the tip of root and shoot

Solution : (Correct Answer: D)

(d) In apical meristem tissue actively dividing cells are present.

150. Plant growth in length is increased by

A) Apical meristem B) Lateral meristem
C) Dermatogen D) Periblem

Solution : (Correct Answer: A)

(a) Apical meristem are found at shoot and root apex. As a result of activity of these meristems plants increases in length.

151. In which part of mitochondria does *ATP* synthesis occur?

- A) F_1
- B) F_0
- C) Cristae
- D) Inner membrane of mitochondria

Solution : (Correct Answer: A)

During the oxidation process (occurs in inner mitochondrial membrane during electron transport system) enormous amount of free energy is released, some of which is utilized by inner membrane sub units of F_1 particles containing three coupling factors and *ATPase* enzyme, in the synthesis of *ATP* molecules

152. Choose the correct combination of *A* and *B* in accordance with the *NCERT* text book.

The *NADH* synthesised in ...*A*... is transferred into the mitochondria and undergoes oxidative ...
B...

- A) *A* – *EMP*; *B* – *carboxylation*
- B) *A* – *ETS*; *B* – *phosphorylation*
- C) *A* – *glycolysis*; *B* – *phosphorylation*
- D) *A* – *TCA cycle*; *B* – *decarboxylation*

Solution : (Correct Answer: C)

The *NADH* synthesised in glycolysis is transferred into the mitochondria and undergoes oxidative phosphorylation

153. In alcohol fermentation

- A) There is no electron donor
- B) Oxygen is the electron acceptor
- C) Triose phosphate is the electron donor, while acetaldehyde is the electron acceptor
- D) Triose phosphate is the electron donor, while pyruvic acid is the electron acceptor

Solution : (Correct Answer: C)

In alcoholic fermentation,

1. *NADH* (formed during conversion of triose-3 phosphate to 3 -phosphoglycerate) is oxidized to NAD^{+} 2. Electrons are accepted by acetaldehyde formed by Decarboxylation of pyruvate

154. How many CO_2 are released by oxidation of 2 molecules of glucose?

- A) 10
- B) 8
- C) 6
- D) 12

Solution : (Correct Answer: D)

155. *RQ* of anaerobic respiration is *B*

- A) 1
- B) 2
- C) Less than 1
- D) Infinite

Solution : (Correct Answer: D)

156. The common immediate source of energy in cellular activity is

- A) *ADP*
- B) *ATP*
- C) *FAD*
- D) *NAD*

Solution : (Correct Answer: B)

It's obvious.

157. How many *ATP* molecule are produced in *Kreb's* cycle

- A) 36
- B) 38
- C) 40
- D) 24

Solution : (Correct Answer: D)

It's obvious.

158. Which one is correct sequence in glycolysis

- A) $G\ 6 - P \rightarrow PEP \rightarrow 3 - PGAL \rightarrow 3 - PGA$
- B) $G\ 6 - P \rightarrow 3 - PGAL \rightarrow 3 - PGA \rightarrow 3 - PEP$
- C) $G\ 6 - P \rightarrow PEP \rightarrow 3 - PGA \rightarrow 3 - PGAL$
- D) $G\ 6 - P \rightarrow 3 - PGA \rightarrow 3 - PGAL \rightarrow PEP$

Solution : (Correct Answer: B)

(b) Glucose -6 - phosphate $\rightarrow$ 3 - phosphoglyceraldehyde $\rightarrow$ 3 - phosphoglyceric acid $\rightarrow$ phosphoenol pyruvate $\rightarrow$ pyruvic acid.

159. *Kreb's* cycle is also known as

- A) Glyoxylate cycle
- B) *EMP* pathway
- C) Citric acid cycle
- D) Glycolate cycle

Solution : (Correct Answer: C)

(a) The first product of *kreb's* cycle is citric acid which is tricarboxylic acid and hence this cycle is also known as citric acid cycle or Tricarboxylic acid cycle.

160. Oxidative phosphorylation occurs during the process of

- A) Protein synthesis
- B) N_2 fixation
- C) Respiration
- D) Transpiration

Solution : (Correct Answer: C)

It's obvious.

161. The tissue of highest respiratory activity is

- A) Meristems
- B) Ground tissue
- C) Phloem
- D) Mechanical tissue

Solution : (Correct Answer: A)

(a) Because metabolic activity rate of meristematic tissue is very high.

162. When *ATP* is converted into *ADP*, it releases

- A) Electricity
- B) Energy
- C) Enzymes
- D) Hormones

Solution : (Correct Answer: B)

(b) The energy is usually released from *ATP* by hydrolysing the terminal phosphate groups.

163. Carbon dioxide liberates during

- A) Ascent of sap
- B) Transpiration
- C) Photosynthesis
- D) Respiration

Solution : (Correct Answer: D)

It's obvious.

164. To a living organism which of the following has the greater amount of available energy per molecule

A) *ATP* B) *ADP* C) *CO₂* D) *H₂O*

Solution : (Correct Answer: A)

It's obvious.

165. *CO₂* is constantly released in respiration yet its level remains fairly uniform because

A) *CO₂* is used as a buffer
B) *CO₂* is utilized in photosynthesis
C) *CO₂* is converted into carbonate rocks
D) *CO₂* undergoes regular photolysis

Solution : (Correct Answer: B)

It's obvious.

166. The enzyme *RuBP* carboxylase

A) Activity occurs in *C₃* and *C₄* plants
B) It present in inner thylakoid membrane
C) Is low temperature sensitive enzyme
D) Shows greater affinity for *O₂* than for *CO₂*

Solution : (Correct Answer: A)

RUBP carboxylase is used in both plants.

167. Which of the following statements is not true regarding the *C₄* plants?

A) They show kranz anatomy
B) Decarboxylation process occurs in bundle sheath cells
C) Granal chloroplast is present in bundle sheath cells.
D) *PEP*case enzyme activity occurs in mesophyll cells

Solution : (Correct Answer: C)

In Bundle sheath cells agranal chloroplast present.

168. *NADPH* is generated through

A) Anaerobic respiration
B) Cyclic photophosphorylation
C) Non-cyclic photophosphorylation
D) Glycolysis

Solution : (Correct Answer: C)

PS I reduces *NADP⁺* to *NADPH* in the stroma by action *NADP* reductase enzyme.

169. In the chloroplast, the stroma lamellae lack the

A) *PS I*, *NADP* reductase
B) *PS II*, *PS I*
C) *NADP* reductase enzyme, *P700*
D) *NADP* reductase, *PS II*

Solution : (Correct Answer: D)

PS II and *NADP* reductase absent in the stroma lamellae of the chloroplast.

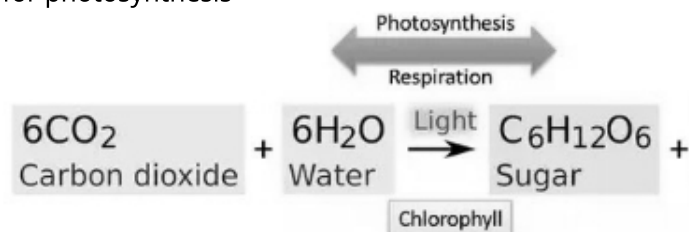
170. Which of the following is/are the raw material for photosynthesis?

I. *H₂O* II. *CO₂* III. Light IV. Chlorophyll
Choose the correct option

A) II, III and IV
B) I and IV
C) I, II and III
D) I, II, III and IV

Solution : (Correct Answer: D)

H₂O, *CO₂*, light, chlorophyll all are the raw material for photosynthesis



171. Which is the evidence to show that *O₂* is released in photosynthesis comes from water

A) Isotopic *O₂* supplied as *H₂O* appears in the *O₂* released in photosynthesis
B) Isolated chloroplast in water releases *O₂* if supplied potassium ferrocyanide or some other reducing agent
C) Photosynthetic bacteria use *H₂S* and *CO₂* to make carbohydrates
D) All the above

Solution : (Correct Answer: D)

It's obvious.

172. Aerobic atmosphere is maintained by

A) Prokaryotes B) Protists
C) Plants D) Fungi

Solution : (Correct Answer: C)

(c) Because *O₂* is released during photosynthesis.

173. What is the unique process which has supported life on this planet

A) *N₂*-fixation B) Photosynthesis
C) Protein synthesis D) Respiration

Solution : (Correct Answer: B)

(b) Because photosynthetic end products are organic materials and *O₂* for respiration.

174. In *C₃* plants, phosphoketopentose epimerase is required for converting

A) Ribose into ribulose
B) Xylulose to ribulose 1, 5 Di *PO₄*
C) Erythrose to xylulose
D) None of the above

Solution : (Correct Answer: B)

It's obvious.

175. No. of carboxylation in *C₄* cycle is/are

- A) 1 B) 2 C) 5 D) 3

Solution : (Correct Answer: B)

(b) Two times carboxylation occur in C_4 cycle first carboxylation is done by phosphoenol pyruvate and second in bundle sheath cell by Ribulose 1, 5 biphosphate.

176. Which of the following photosynthetic bacteria have both $PS - I$ and $PS - II$

- A) Green sulphur bacteria
B) Purple sulphur bacteria
C) Cyanobacteria
D) Purple non-sulphur bacteria

Solution : (Correct Answer: C)

It's obvious.

177. The detectable end product of photosynthesis is

- A) Glucose
B) Fructose 1, 6 diphosphate
C) Ribulose phosphate
D) Starch

Solution : (Correct Answer: D)

(d) Starch is the detectable end product of photosynthesis because glucose is converted and stored in the form of starch.

178. Importance of photosynthesis is due to

- A) Synthesis of food
B) Purification of atmosphere
C) Provided vast resources of energy
D) All the above

Solution : (Correct Answer: D)

It's obvious.

179. Photosynthesis is a

- A) Exothermic process
B) Exergonic process
C) Anabolic process
D) Catabolic process

Solution : (Correct Answer: C)

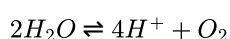
(c) Photosynthesis is an anabolilc and endothermic reaction. It is a mechanism of synthesis of food.

180. During formation of glucose which of the following is obtained from water

- A) Oxygen
B) Hydrogen
C) Both (a) and (b)
D) None of the above

Solution : (Correct Answer: B)

(b) During formation of glucose water is oxidised and hydrogen obtain from water.





Mitoprep

Sapno Ki Udaan